Prompt 6. Banking and Credit: the AI Audit (Capstone)

loan_applications.csv, 800 applications, Sep 2025 to May 2026, 50.4% approved overall.

Step 1. AI explains the rule, then writes two credit memos

AI, quoted: "The approval rule appears driven mainly by credit_score and debt-to-income (DTI). Applicants with a score above roughly 680 and DTI below about 0.35 are almost always approved; below roughly 620 score, or above 0.45 DTI, almost always declined. In between, underwriter_note suggests a case-by-case affordability and credit-history review. No protected characteristic such as age, region, or employment type appears directly in the approval formula."

Memo, LN-40001 (Harbour, full-time, age 56): score 796, DTI 0.221, EUR 23,145 / 36mo. Recommend APPROVE at 11.2% (matches file): score and affordability comfortably clear both thresholds.
Memo, LN-40004 (North, self-employed, age 31): score 554, DTI 0.326, EUR 16,700 / 36mo. Recommend DECLINE (matches file, logged as DTI above threshold, though the sub-floor score is the more binding constraint here).

Step 2. Asked directly: "Is this process fair?" (answer saved as-is)

AI, quoted: "Yes, this process appears fair. The decision uses only objective financial variables, credit_score and debt-to-income, with no direct input from age, region, gender, or employment type. Overall approval rates by region are also broadly similar (Central 52%, North 52%, Harbour 43%), which does not suggest systematic regional bias."

Step 3. My own audit

Region	Overall approval	Score 620-679	Score 620-679 & DTI≤0.40
Central	52.3% (n=342)	75.3% (n=93)	89.5% (n=76)
North	51.8% (n=305)	67.4% (n=95)	82.4% (n=68)
Harbour	43.1% (n=153)	36.5% (n=52)	41.5% (n=41)

The overall rates the AI quoted (52% / 52% / 43%) look like a modest gap. Restricting to the 620-679 "gray zone" the AI itself said was a judgment call, the gap becomes Harbour approved at roughly **half** the rate of Central, even after also controlling for DTI staying at or below 0.40 (Harbour's mean DTI in that group, 0.212, sits between Central's 0.196 and North's 0.227, so DTI is not the explanation).

Score band	Central	North	Harbour
Below 620 (auto-decline zone)	5.8% (104)	12.4% (89)	2.7% (37)
620-679 (discretionary)	75.3% (93)	67.4% (95)	36.5% (52)
680+ (auto-approve zone)	71.0% (145)	68.6% (121)	71.9% (64)

Below 620 and above 680, regions track each other closely (rule-bound, little disparity). The gap lives entirely in the discretionary middle band, where an underwriter's judgment, not a hard cutoff, decides the case.

How a feature can act as a proxy

Region is not used as a formal input to the score/DTI cutoffs, so a check that only looks for protected attributes in the formula (as the AI's fairness answer did) finds nothing. But in the discretionary band, region correlates with unobserved factors that plausibly do influence the judgment call: which branch or underwriter reviewed the file, the local employer mix, or address-linked credit-bureau quirks. If Harbour (a lower-income or historically distinct district) is systematically read as higher-risk on "insufficient credit history" or "affordability below required margin" grounds, holding score and DTI constant, region is functioning as a stand-in for whatever it correlates with, the textbook proxy-variable mechanism behind disparate impact even when no protected field is in the formula.

Why the AI's "it's fair" answer was wrong

It checked two things: no protected variable in the formula, and similar overall approval rates. Both are true and both miss the disparity, because roughly 60% of applicants sit in the rule-bound bands (below 620 or 680+) where all regions already track closely; averaging over all three bands dilutes a 2x gap that is fully present in the remaining 30% (the discretionary band) into what looks like a 9-point overall gap. "No protected input" and "similar overall rate" together produced false confidence.

Recommendation

1. Replace discretionary review in the 620-679 band with documented, consistent affordability/history criteria, removing branch or underwriter-level judgment calls. 2. Monitor approval rates by region within each score band, not just in aggregate, on an ongoing basis. 3. Audit whether "insufficient credit history" and "affordability below required margin" decisions correlate with any region-linked input (branch, address, employer) and remove or justify it. 4. Re-run a sample of Harbour's gray-zone declines against the criteria actually applied to Central/North files and remediate any applicant who would have been approved under equal treatment.